

Event Studies

Chris Conlon

Saturday 8th November, 2025

Applied Econometrics

Event Studies

Recall: Fundamental Problem of Causal Inference

Fundamental Problem of Causal Inference

- ▶ We don't observe the **counterfactual** $Y_i(1 - D_i)$.
- ▶ For a single individual we either observe $Y_i(1)$ **or** $Y_i(0)$ but never both!
- ▶ Matching says: Find someone with $1 - D_i$ but the same X_i (or **re-weight** your sample)
 - Assume all selection into treatment is based only on observed X_i
 - Ignore **selection on unobservables**

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

i	$Y_i(1)$	$Y_i(0)$	D_i	X_i
1	1	?	1	3.2
2	0	?	1	1.7
3	?	0	0	2.5
		$\vdots$		
n	?	1	0	-1.5

How do we construct the counterfactual?

We observe $Y_i(D_i)$ but not $Y_i(1 - D_i)$, how do we estimate it?

Event Study: Compare $Y_{t+k}(1)$ to $Y_{t-j}(0)$ before and after an event at t .

Matching: Compare $Y_i(1)$ to $Y_j(0)$ for $i \neq j$ but where $X_i = X_j$.

Difference in Difference: Compare $Y_{i,t}(1) - Y_{i,t-1}(0)$ to $Y_{j,t}(0) - Y_{j,t-1}(0)$ for some $j \neq i$.

Synthetic Control: Compare $Y_{i,t}(1) - Y_{i,t-1}(0)$ to $\bar{Y}_{-i,t}(0) - \bar{Y}_{-i,t-1}(0)$ for some average over $j \neq i$ so that $X_{i,t} \approx \bar{X}_{-i,t}$.